Revisiting “Remarks”: the learning of gerundive and derived nominals

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Descriptive and explanatory adequacy

- Since Chomsky 1965, we have evaluated linguistic theories in two ways: **descriptive** and **explanatory adequacy**.
  - Descriptive adequacy: does the theory capture the empirical facts?
  - Explanatory adequacy: can the theory be learned by a child?
- The usual way of doing things: descriptive → explanatory
- A new approach: explanatory → descriptive

Today

- Today's goal: use this new approach to understand English nominalizations
- I take Chomsky 1970 "Remarks on nominalization" as my starting point.

Chomsky 1970

The central focus of Chomsky 1970: gerundive vs. derived nominals

(1) a. John refused the offer. (clause)

b. John's refusing the offer (gerundive nominal)

c. John's refusal of the offer (derived nominal)

Chomsky 1970

The most striking differences (Chomsky 1970: 15–17):

1. Productivity
   - Gerundive nominals are productive; derived nominals are not.
2. Semantic relation between the nominal and proposition
   - Gerundive nominals preserve vP semantics: *John's impressing the teacher*
   - Derived nominals need not: *John's impression of the teacher*
3. Internal structure of the nominal
   - Gerundive nominals have vP structure: *His repeatedly/*repeated interrupting the speaker*
   - Derived nominals have NP structure: *His *repeatedly/repeated interruption (*of) the speaker*

Note on today's talk

- A huge literature on English nominalizations: Abney 1987, Grimshaw 1990, Marantz 1997, Fu et al. 2001, Alexiadou 2001, Borer 2003, 2014, Alexiadou & Borer 2020, a.o.
- The focus today is on deriving a theory that achieves explanatory adequacy.
- We will only show descriptive adequacy to the extent of Chomsky 1970.
  - See appendix for further empirical coverage, including
  - mixed nominals like *Mary's reading of the book*
  - and "acc-ing gerunds" (Horn 1975, Abney 1987) like *Him refusing the offer*

Roadmap

1. Introduction
2. The theory of learning
3. Learning gerundive nominals
4. Learning derived nominals
5. Conclusion

The theory of learning

The notion of productivity

Core to our theory of learning is the notion of **productivity**.

- Productive rules[1]
  - V + [PAST] → V-*ed* (productive over V)
- Unproductive rules
  - think + [PAST] → thought (unproductive; memorized for each verb)

[1] I use "rule" as a general term to mean anything a child learns.

The learning of a single rule

How are rules learned?

- By default, rules are memorized (i.e. unproductive)
  - A child starts by memorizing *run-ran*, *think-thought*,
  - ... as well as *laugh-laughed*, *smile-smiled*
- The child **generalizes** the rule as productive over some domain, iff they receive **sufficient evidence**.
  - Once the child hears enough verbs in *-ed*, they generalize the rule as productive.

What is sufficient evidence?

(2) The Tolerance Principle (Yang 2016)

Let a rule R be defined over some domain of size N. Let M be the number of examples[a] attested in the input that support R. Define

$$\theta_N = \frac{N}{\ln N}.$$

Then, R **is productive over the domain if and only if**

$$M \geq N - \theta_N.$$

[a] Number of types, not tokens

The learning of a grammar

Now we know how a rule is learned... How is a grammar learned?

- Over the course of development, the child learns rules incrementally.
- The resulting **set of interdependent rules** is the adult grammar.
- Thus, learning a grammar = learning the individual rules that comprise the grammar.

The practical logic: evaluating learnability

- To evaluate the learnability of a rule:
  1. Look at a child's input data
  2. If unproductive rule:
     - As long as it occurs in the input, it can be learned.
  3. If productive rule:
     - It can be learned iff the input provides sufficient evidence for productivity via TP.
- To evaluate the learnability of a grammar, evaluate the individual rules.

The practical logic: evaluating learnability

- As a proxy for the child's input, I use CHILDES (MacWhinney 2000), 6.8-million tokens of English child-directed speech.
- As a proxy for the child's vocabulary of verbs, I use the most frequent 100 verbs in my data.
  - We focus on verbs because we are mostly concerned with rules stated over verbs.
  - Children learn the core rules of language at an early age, on a small vocabulary of frequent words (Goodman et al. 2008).[2] Thus we use a frequency-trimmed list of words as their vocabulary (Belth 2023, Kodner 2023, Thoms et al. 2025, Lu et al. 2026).

[2] For example, *-ing* is the first morpheme children learn, well before age 2 (Brown 1973, Mintz 2013).

Learning gerundive nominals

Setting up the learning problem

Let's first consider the learning problem children are faced with...

- Gerundive nominals are rare.
  - Gerundive nominals: ~220 tokens
  - Gerundive nominals in top 100: ~66 tokens, 26 verb types
  - Gerundive nominals in top 100 with direct object: 9 tokens, 7 verb types[3]
- This input cannot support any productive rule! Yet gerundive nominals are clearly productive.
- So how do children learn them?
  - A **poverty of the stimulus** problem!

[3] Children hear ~5 million words per year, so these numbers are around 1 gerundive nominal every three days, 5 with direct objects every year.

Setting up the learning problem

- A path forward if we consider the broader input...
- An abundance of examples with *-ing*
  - *-ing*: 105k tokens. 98% participles, 2% gerunds.
  - *-ing* in top 100: 80k tokens, 99 verb types.

(3)
a. you're [going to hurt me] (progressive)
b. he came towards me [waving] (secondary predication)
c. I enjoyed [reading your book] (complement to *enjoy*)
d. you were thinking about [being a teacher] (complement to *about*)
e. let's tell them about your [breaking your finger] (gerundive nominal)

If only the child can make use of all these data in learning gerundive nominals...

Summary of my analysis

What is learned? Two rules:

1. The category ingP
2. Selection of ingP by possessive heads

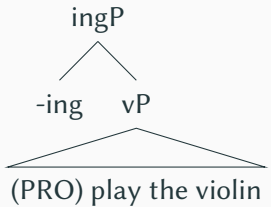
Together, these two rules generate gerundive nominals.[4]

[4] Plus prior knowledge (e.g. basic vocabulary and basic categories like VP, vP, NP, DP).

Rule 1: the category ingP

- A consistent unit recurs across varying environments
  - Internally: *-ing* plus vP structure
  - Externally: various syntactic environments, distinct distribution from vP
- This motivates a syntactic category: ingP

(4)



Rule 1: the category ingP

- On the basis of (3), the child memorizes lexical rules
  - *-ing* selects *go, wave, say, read, be*, etc.
- But, looking at the broader input, the child can generalize the rule as productive.
  - Of the top 100 verbs, $M = 99$ are attested with *-ing*. The threshold is 79 ($100 - \theta_{100}$).
- Furthermore, this is productivity over vP, not just V.
  - Of the 83 transitive verbs in top 100, $M = 80$ are attested with *-ing* plus direct object. The threshold is 65 ($83 - \theta_{83}$).

Rule 1: the category ingP

$[_{\text{ingP}}$ -ing vP] (productive over vP)

Rule 2: D_{poss} selects ingP

- Once the child learns ingP, they can learn further rules about ingPs.
- One such rule: D_{poss} selects ingP[5]
  - Of the 7 elements of D_{poss} (*'s*, *my*, *his*, ...), all $M = 7$ are attested with ingPs.

Rule 2: D_{poss} selects ingP

$[_{DP}\ D_{poss}\ \ \text{ingP}]$ (productive over D_{poss})

[5] Crucially, the child only needs to learn a selectional rule between D_{poss} and *-ing* (the head of ingP). Rule 1 takes care of the internal productivity of ingPs.

Consequences of the rules

- Rule 1 and Rule 2 together generate gerundive nominals:[6]

(5) a. $[_{vP}$ play the violin] (prior knowledge of vP)

b. $[_{ingP}$ -ing $[_{vP}$ play the violin]] (Rule 1: the category ingP)

c. $[_{D'}$'s $[_{ingP}$ playing the violin]] (Rule 2: D_{poss} selects ingP)

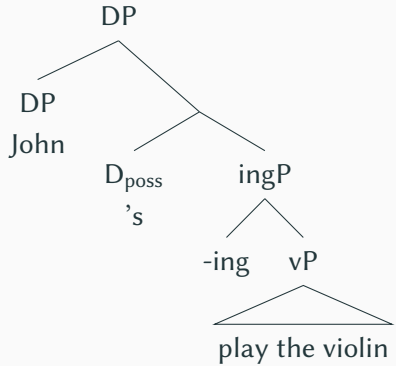
d. $[_{DP}$ John's playing the violin] (prior knowledge of possessives)

[6] I remain neutral on the treatment of the subject in an ingP.

Consequences of the rules

- These rules generate the structure of gerundive nominals as follows:[7]

(6)



[7] This converges on standard analyses of gerundive nominals as containing VP/vP, with an outer DP shell (e.g. Abney 1987, Kratzer 1996, Marantz 1997).

Consequences of the rules

- Descriptive adequacy of the analysis:[8]

Part of analysis	Desired empirical prediction
ingP contains vP	**vP syntax and semantics**
ingP contains vP	Can have overt subjects: "acc-ing gerunds"
No productivity over D	No other Ds occur in gerundive nominals: **the playing the violin*
ingP $\neq$ DP	Distributional differences between ingP and DP
ingP as a category	Unified treatment of *-ing* in different environments

[8] See appendix for more details.

Interim summary

- We posed the learning of gerundive nominals as a poverty of the stimulus problem.
- We resolved this via appropriate decomposition of the learning problem.
  - Children can use abundant examples with *-ing* to learn a productive rule over vP (harder to learn, but more data)
  - Then use the subset of examples with gerundive nominals to learn a productive rule over D_{poss} (less data, but easier to learn)

  $\rightarrow$ explanatory adequacy ✓
- The two rules produce the structure for gerundive nominals in (6), which makes desired empirical predictions.

  $\rightarrow$ descriptive adequacy ✓

Learning derived nominals

What is not learned

- The child's input data around derived nominals looks very different...
- No evidence for productivity over verbs
  - The most frequent nominalizers are *-tion*, *-ment*, and *-al*. They each occur only with 1 verb in the top 100 (*fixation, movement,* and *trial*). This is far below the productivity threshold.
- No evidence for vP structure
  - Derived nouns never occur with material diagnostic of vP structure, e.g. direct objects.

What is learned

- Instead, what is learned are individual words like *destruction*[9]
  - This is just ordinary word learning.
- Derived nominals and their properties follow.

[9] I use "derived noun" to refer to words like *destruction*.

Rule 3: Lexical entry for *destruction*

- Based on the distribution of the word and how it is used, the child learns its category and meaning.

Rule 3: Lexical entry for *destruction*

a) Category N (memorized for each word)

b) $[\![\textit{destruction}]\!] = \lambda e[\text{destroy}(e)]$ (memorized for each word)

Consequences of the rules

- Rule 3 generates derived nominals:

(7) a. $[_{N}$ destruction$]$ (Rule 3a: category of *destruction*)

b. $[_{NP}\ [_{N}$ destruction$]\ [_{PP}$ of the city$]]$ (prior knowledge of NPs)

c. $[_{DP}$ The army's $[_{NP}$ destruction of the city$]]$ (prior knowledge of poss.)

(8) $[\![$*the army's destruction of the city*$]\!]$
$= \lambda e[\text{destroy}(e)$ (Rule 3b: meaning of *destruction*)
$\wedge R(e, \text{the city})$ (prior knowledge of *of*)
$\wedge R'(e, \text{the army})]$ (prior knowledge of poss.)

Consequences of the rules

- Descriptive adequacy of the analysis:

Part of analysis	Desired empirical prediction
Derived nouns are Ns	**Derived nominals are NPs, no vP syntax or semantics**
Derived nouns are event predicates	Derived nominals are interpreted as complex events (e.g. event participants, aspect)

Conclusion

Summary

- I have shown how gerundive and derived nominals can be learned, given a concrete theory of learning.
- This learning story produces syntactic proposals with concrete empirical predictions.
- We can also understand the differences between gerundive and derived nominals noted by Chomsky (1970) as this result of learning.

Zooming out

- This case study illustrates a general approach:
  1. Propose a theory that achieves explanatory adequacy, by showing how it can be learned.
  2. Then, show that it achieves descriptive adequacy.
- Beyond the methodological point, this way of thinking also integrates our theoretical understanding of language with the psychologically grounded view of language as a product of learning.
  - This is how we connect theoretical linguistics to broader work on language.

Notes

- I am indebted to Dan Brodkin, Alexander Hamo, Muhammed Ileri, Julie Legate, Charles Yang, and the participants of Penn Syntax Lab for extensive feedback on earlier versions of the talk/manuscript.

- Manuscript and slides available at: liankwang.github.io.
- Appendices on:
  - Further empirical predictions
  - Mixed nominals and acc-ing gerunds
  - Morphological relationship between derived nouns and verbs

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Appendix

Empirical consequences

Gerundive nominals are only nominal at the DP level. They do not contain NPs.

- Thus we understand why gerundive nominals cannot appear with just any D

(9) a. *The playing the violin (by John)

b. The playing of the violin (by John)

Empirical consequences

ingPs have a unified **internal** syntax and semantics.[10] Their various uses differ in the **external** environment, including how the subject is interpreted in that environment.

- Gerundive nominals are possessives: spec,D_{poss} identified as open argument in the relation introduced by its complement
- Control structures: PRO in ingP
- Matrix subject: generic PRO
- Adjunction structures: nominal modification or secondary predication

[10] Additional things need to be said for where the aspectual meaning of progressives comes from.

Empirical consequences

Rules involved in other ingP uses:

(10)	a.	you're [going to hurt me]	$[_{VP}$ be ingP$]$
	b.	I enjoyed [reading your book]	$[_{VP}$ enjoy ingP$]$
	c.	you were thinking about [being a teacher]	$[_{PP}$ about ingP$]$
	d.	he came towards me [waving]	$[_{VP}$ VP ingP$]$
	e.	all those fish eyes [looking at me]	$[_{NP}$ NP ingP$]$

Empirical consequences

"Acc-ing gerunds" are just ingPs with overt subjects:[11]

- Shared distribution...

(11) a. John enjoys [$_{\text{ingP}}$ (Bill) playing the violin].

b. John left the room without [$_{\text{ingP}}$ (Bill) seeing the teacher].

c. [$_{\text{ingP}}$ (Bill) playing the violin] delights John.

- with differences due to subject realization:

(12) a. The [$_{\text{NP}}$ [$_{\text{NP}}$ student] [$_{\text{ingP}}$ (*Bill) playing violin]] delighted John.

b. John [$_{\text{VP}}$ [$_{\text{VP}}$ left the room] [$_{\text{ingP}}$ (*Bill) cursing his teacher]]

c. John saw [$_{\text{ingP}}$ *(Bill) playing the violin]

[11] Similar treatment as Pires (2007) and Reuland (1983)

Empirical consequences

ingPs are not DPs!

- Even though they overlap, their distributions are different:

(13) a. [$_{\text{ingP}}$ (The clerk) having left the store], John ate a cookie.

b. *[$_{\text{DP}}$ The clerk's having left the store], Mary stole a cookie.

(14) a. I expected [$_{*\text{ingP/DP}}$ *him/his leaving early].

b. The scandal necessitated [$_{*\text{ingP/DP}}$ *him/his leaving the company].

c. I recommended [$_{*\text{ingP/DP}}$ *him/his leaving immediately]

Empirical consequences

ingPs are not DPs.

- ingPs do not display some properties of DPs:

(15) Plural agreement with the verb

a. [$_{DP}$ John's coming] and [$_{DP}$ Mary's leaving] *bothers/bother me.

b. [$_{ingP}$ John coming] and [$_{ingP}$ Mary leaving] bothers/*bother me.

(16) Extraction out of DP/ingP

a. *The city that we remember [$_{DP}$ his describing t].

b. The city that we remember [$_{ingP}$ him describing t].

(Abney 1987: 175–178)

Learning mixed nominals

- What about mixed nominals like *John's repeated opening of the door*?
- Learning mixed nominals has two components:
  1. Productivity of *-ing* over verbs (as with ingPs)
  2. V-*ing* occurs in the distribution of Ns $\rightarrow$ learned as category N
- Thus mixed nominals are derived nominals, whose "derived noun" is productively formed from a verb.

The morphological relationship bewteen *destroy* and *destruction*

- We didn't make use of the morphological relationship between a derived noun and its verb.
- This is learnable (Ileri & Hamo 2026), but not necessary.
- The fact that CEN interpretations are tied to a morphogical relation between the derived noun and a verb can be understood via independent reasons in English (diachrony and productive conversion). This association does not hold for some languages, e.g. French (Huyghe et al. 2017).

See manuscript for more discussion.